Gyumin Park

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Research Interests

- Continual and few-shot learning, deciding which parts of a pretrained model change and which stay fixed
- Robust adaptation of vision foundation models to new domains and conditions
- Physical understanding and world models, reading state-space models as dynamical systems

Education

Kookmin University, Seoul, Republic of Korea Mar 2020 – Aug 2026
B.S. in Nano and Electronic Physics; B.S. in Computer Engineering (second major: Software)
GPA 4.15 / 4.5 (96.5 / 100 on the university's percentage scale)

Research Experience

Research Intern, Center for Artificial Intelligence, AI & Robotics Institute, KIST Jul 2026 – Present
Automatic labeling of traffic CCTV footage with foundation models, from object boxes to vehicle attributes ongoing

- Adapted SAM3 to detect six traffic object classes with class-wise LoRA modules selected by the text query, raising box AP50 from 0.540 (zero-shot) to 0.627.
- Used a VLM to catch false detections and misclassifications from SAM3, and to caption vehicle attributes such as color and type from whole frames for real-time labeling.

Undergraduate Researcher, HCI Lab, Kookmin University Advisor: Prof. Sangmin Yoon Jan 2024 – Aug 2026
SS-MoE: few-shot class-incremental learning with selective-scan experts first author; in preparation

- Identified that prompt-, prototype- and branch-based FSCIL methods decide adaptation at the level of prompts, global prototypes or whole branches, and formulated adaptation as an input-conditioned routing problem.
- Designed SS-MoE, which freezes a VMamba backbone after the base session, uses an ETF classifier fixed from the start, and adapts only a mixture-of-experts neck of four-direction selective-scan experts routed by SS2D scan states.
- Evaluated on CUB-200, mini-ImageNet and CIFAR-100. On CUB-200, decomposed the accuracy one component at a time: the routed neck added 6.08 points over a linear neck, more than the classifier and loss combined (4.37) or the higher input resolution (2.46).
- Under Top-2 routing, showed that the direction-aware router keeps experts evenly used without a load-balancing loss (std 0.44% vs 11.44% for a linear router) and beats it by 4.14 points on novel classes.
- On CUB-200, swapping the selective-scan experts for MLP experts lowered accuracy in every session, by 3.56 points on average, even though the MLP experts had more parameters.

SpanInit: how long the states of Mamba-based vision backbones keep information first author; under review

- Set out to tune frozen Mamba-based vision backbones the way prompt-based methods tune frozen ViTs. One way is to add new state dimensions, as in Additional-scan, but each one needs an initial decay timescale.
- Showed that relative decay timescales within a channel depend only on the state matrix A and read them from nine public checkpoints without data. ImageNet-1k-only ones with several states per channel had near-collapsed timescales, while ImageNet-21k-exposed ones spanned more than three orders of magnitude.
- Proposed SpanInit, which places the added states in the largest uncovered gaps of the scan-relevant timescale range, with no extra trainable parameters or inference cost beyond the added states.
- Evaluated on VTAB-1k with MambaVision-B/L over three seeds: against Additional-scan's default initialization, +0.39 / +0.59 points on ImageNet-21k checkpoints and -0.38 / -0.21 on ImageNet-1k-only ones, a contrast significant at $p < 0.001$ on MambaVision-B. This motivates initializing added states to fit each checkpoint's timescale geometry.

Manuscripts

Under review

[1] **Park, G.**, & Yoon, S. (2026). Pretrained Mamba-based vision backbones can exhibit collapsed timescale banks: Checkpoint-aware state expansion with SpanInit. Manuscript under review.

In preparation

[2] **Park, G.**, & Yoon, S. (2026). Selective scanning with mixture-of-experts for few-shot class-incremental learning. Manuscript in preparation for journal submission.

Projects

R&Dix, Inter-University Student Research Club Jun 2025 – Dec 2025
Driving with imperfect human intervention team project; implemented the extensions and ran all experiments

- Extended an existing safe human-in-the-loop reinforcement learning framework for highway driving in the SMARTS simulator, built on a soft actor-critic agent that shares control with a human.
- Connected an LLM that suggests a high-level driving action at each step, and simulated imperfect human interventions.
- Trained with simulated imperfect interventions, the agent succeeded in 0/10 evaluation episodes, with seven off-road failures, against 8/10 for an agent trained without them (one run each).

Industry Experience

Research Intern, QSimPlus Fall 2025

- Optimized quantum key distribution operating parameters with genetic algorithms.
- Trained an FT-Transformer that predicts optimized parameters and the secret key rate from device conditions, with a 2.06% mean error on the key rate, as a faster alternative to genetic-algorithm search in resource-constrained settings.

Awards and Honors

Top Prize, UNITEF AI Application Proficiency Program; selected for the Singapore Global Training Program 2025
Summer Research Internship Scholarship, Kookmin University 2025
Undergraduate Internship Support Scholarship, Kookmin University 2025
National Undergraduate Scholarship, Korea Student Aid Foundation 2025
Insong Foundation Fellowship, Insong Cultural Foundation 2024
Academic Excellence Scholarship, Kookmin University 2024

Technical Skills

Programming: Python, LaTeX, Git
Frameworks: PyTorch, TensorFlow, OpenCV
Architectures: state-space models (Mamba, VMamba, MambaVision, VideoMamba), ViTs, CNNs, RNNs, generative models (GANs, VAEs, diffusion), world models (JEPA, Dreamer)
Environment: Linux (Ubuntu), Windows, CUDA

Languages and Test Scores

Korean: native
English: TOEFL iBT 5.5 / 6, equivalent to 110 / 120 on the previous scale (2026)
Japanese: JLPT N2 (2023)

Training Programs

AI Application Proficiency Program, UNITEF Summer 2025

- Built a chemical-process recovery system with GAN data augmentation, CNN-LSTM anomaly detection, sequence-to-sequence recovery and LLM-based summaries of process states.

LG Aimers: AI Professional Program, LG AI Research Summer 2025

- Completed a machine learning and deep learning curriculum, and built a predictive model on large-scale industrial data in the program's hackathon.

Relevant Coursework

Computer Vision (A+), Artificial Intelligence (A+), Practices in Visual Computing (A+), Computer Graphics (A+), Advanced Topics in Computer Science (A+), Differential Equations and Linear Algebra (A+), Data Structure (A+), Computational and Mathematical Physics (A+), Mechanics I (A) and II (A+), Quantum Mechanics I (A+)